

Saint Petersburg State
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End-to-end Sample ML Project course

Report 4: Model Deployment Preparation

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Giskard Vulnerability Scan

The best model (Random Forest Regressor) was wrapped using Giskard and scanned for potential issues on the test dataset.

Scan Configuration

- Model: Random Forest Regressor (n_estimators=100, max_depth=20)
- Dataset: X_test with sellingprice labels (9,703 rows)
- Scan type: Performance bias detection across feature slices

Scan Results

Giskard completed the scan in 18 seconds and found 15 potential issues:

- Performance bias detected on certain feature slices (e.g. specific hash values for model/seller)
- MAE varies slightly across slices but differences are small ($Dm < 0.1$)
- No critical data leakage or label issues detected
- Global MAE = 2,097 - consistent with model training results

The full Giskard HTML report is saved to reports/giskard_report.html

Model Saving Methods Comparison

Three model serialization methods were tested and compared using %%timeit (7 runs, 10 loops each):

Method	Inference Time	File Size	Use Case
Pickle	86.9 ms (+/-1.18ms)	~112 MB	Standard Python serialization
Joblib	86.5 ms (+/-725us)	~112 MB	Optimized for numpy arrays
ONNX	101 ms (+/-2.15ms)	~61 MB	Cross-platform production format

Analysis

- Pickle and Joblib show almost identical inference speed (~86.5 ms)
- Joblib is marginally faster (0.4 ms) and is the recommended format for sklearn models
- ONNX is slower (101 ms) for this model due to high feature count (167 columns)

- ONNX has smaller file size (61 MB vs 112 MB) due to better compression
- For production deployment with cross-platform requirements, ONNX is preferred despite slightly slower inference

Recommendation: Joblib for Python-only environments, ONNX for production/cross-platform deployment.